**July2022 MTH208 EQP Final Solution (For Student's Reference)**

1a)

$$\begin{pmatrix} -2 & 0 & 0 & 0 & 0 \\ 1 & -2 & 0 & 0 & 0 \\ 0 & 0 & -2 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 3 \end{pmatrix}$$

1b) 4

2a)

$$Q = \begin{pmatrix} 1 & 0 & 0 \\ -2/3 & 1 & 0 \\ 13/8 & 1/16 & 1 \end{pmatrix}$$

The rank of $f$ is 3, and its signature is $(1, 2, 0)$.

2b(i)

$$\begin{pmatrix} 3 & -4 \\ 2 & 0 \end{pmatrix}$$

2b(ii)

$$\begin{pmatrix} 3 & -46 \\ 20 & 16 \end{pmatrix}$$

2b(iii)

Such a basis $\gamma$ does not exist.

3a)

$V = W_1 \oplus W_2$, where a basis of $W_1$ is $\{(-3,2,3)^T\}$ and a basis of $W_2$ is $\{(2,1,0)^T, (-2,0,1)^T\}$.

3b)

$$A^{4n} = \begin{pmatrix} -2+3(16^n) & 6-6(16^n) & -6+6(16^n) \\ 2-2(16^n) & -3+4(16^n) & 4-4(16^n) \\ 3-3(16^n) & -6+6(16^n) & 7-6(16^n) \end{pmatrix}.$$

3c)

$$[T]_B = \begin{pmatrix} 73 & 103 & 140 \\ 30 & 47 & 60 \\ -61 & -88 & -118 \end{pmatrix}$$

4a)

$\mathcal{C} = \{v_1, v_2, v_3\}$ is an orthogonal basis of $W$

where

$v_1 = w_1$

$v_2 = -w_1 + w_2$

$v_3 = -6w_1 + w_2 + w_3$

5a)

$$Q = \frac{1}{\sqrt{10}} \begin{pmatrix} 1 & -3 \\ 3 & 1 \end{pmatrix}$$

$$D = \begin{pmatrix} 36 & 0 \\ 0 & 16 \end{pmatrix}$$

5b)

$A = UP$, where

$$P = \begin{pmatrix} 21/5 & 3/5 \\ 3/5 & 29/5 \end{pmatrix}$$

$$U = \begin{pmatrix} 4/5 & -3/5 \\ 3/5 & 4/5 \end{pmatrix}$$

5c)

$A = U\sum V^*$, where

$$U = \frac{1}{\sqrt{10}}\begin{pmatrix} 1 & -3 \\ 3 & 1 \end{pmatrix}$$

$$\Sigma = \begin{pmatrix} 6 & 0 \\ 0 & 4 \end{pmatrix}$$

$$V = \frac{1}{\sqrt{10}}\begin{pmatrix} 1 & -3 \\ 3 & 1 \end{pmatrix}$$

5d)

$A$ and $B$ have different trace, hence they cannot be similar.

6a)

A basis of $W^0$ is given by $\{f_1, f_2\}$, where

$f_1((x_1, x_2, x_3, x_4)^T) = -8x_1 - 3x_2 + x_3$.

$f_2((x_1, x_2, x_3, x_4)^T) = (34/3)x_1 + (11/3)x_2 + x_4$.

6b(ii)

$g_1(w_1) = -7/13$ , $g_2(w_1) = -2/13$ , $g_1(w_2) = -4/13$ , $g_2(w_2) = -3/13$.